1 Month Internship Ethical Hacking Tasks – Week 1 Complete (Basic)

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Ethical Hacking Tasks

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DAY 1

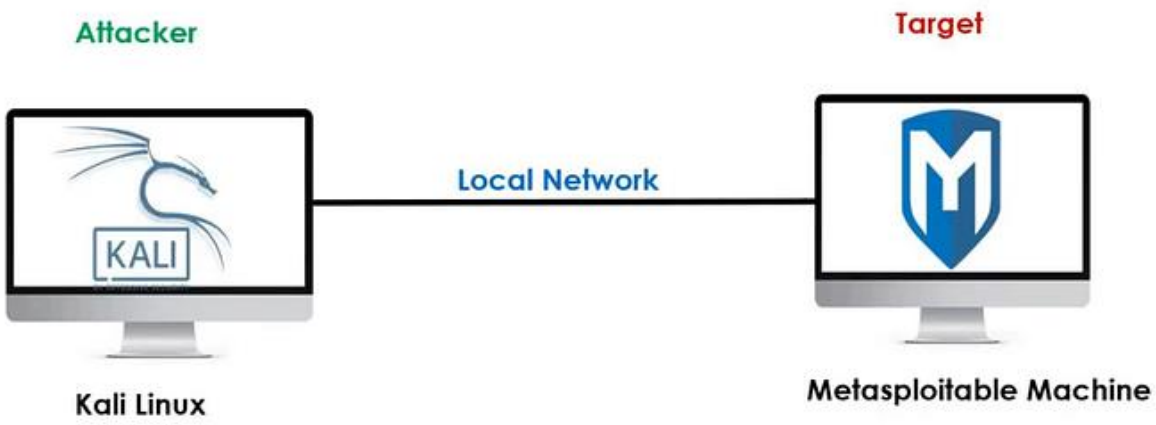
Setting Up the Ultimate Hacking Lab

Introduction

To kick off the **YoungDevInterns_Ethical_Hacking_Internship**, the first day is all about building a safe, legal, and powerful environment. In the world of cybersecurity, you cannot practice hacking on real-world websites or live networks without explicit permission. Therefore, we build a "sandbox" an isolated digital playground where we can learn and fail safely.

1. Objective of the Day

The primary goal is to transform a standard computer into a professional penetration testing station. This involves setting up a **Hypervisor** to host multiple operating systems and creating a private network so the "Attacker" machine can interact with the "Victim" machine without any risk to the outside world.



2. Tools & Software Used

- **VirtualBox:** A powerful, free virtualization tool that allows us to run multiple "Virtual Machines" (VMs) simultaneously.
- **Kali Linux:** The industry-standard operating system for hackers, pre-loaded with over 600 security tools.
- **Metasploitable2:** An intentionally vulnerable Linux-based virtual machine used specifically for security testing.

3. Step-by-Step Implementation

a) Virtualization Setup

First, I installed **Oracle VM VirtualBox**. This software acts as the hardware layer for our virtual lab. It allows us to allocate specific amounts of RAM, CPU, and Disk space to our guest operating systems.

b) Installing the Attacker (Kali Linux)

I downloaded the pre-configured Virtual Machine image from the official Kali website.

- I imported the `.ova` file into VirtualBox.
- I adjusted the settings to provide **2GB of RAM** and **2 CPU cores** to ensure the OS runs smoothly during heavy scanning tasks.

c) Installing the Victim (Metasploitable2)

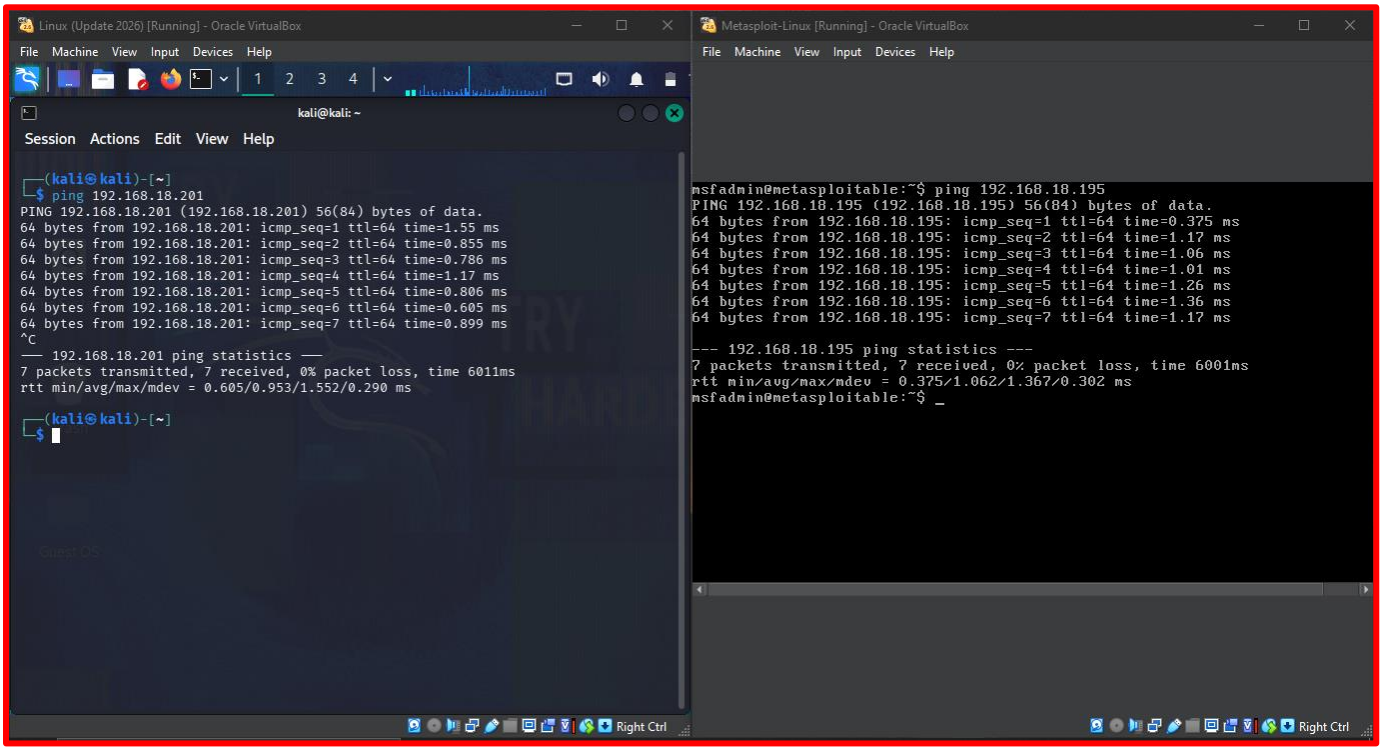
Since Metasploitable2 is a target machine, I set it up using its virtual disk file (`.vmdk`).

- I created a new "Linux" machine in VirtualBox.
- I selected "Use an existing virtual hard disk file" and pointed it to the Metasploitable file.

4. Network Configuration & Security

This is the most critical step for any ethical hacker.

- I configured both machines to use a **Host-Only Network** (or a private NAT Network).
- **Reason:** This ensures the two machines can "talk" to each other, but my hacking activity stays completely isolated from my local home Wi-Fi and the public internet. This prevents accidental damage or legal issues.



5. What I Did Today

1. **Environment Build:** Successfully installed the Hypervisor (VirtualBox).
2. **OS Deployment:** Deployed Kali Linux as my primary hacking platform.
3. **Target Setup:** Set up Metasploitable2 to serve as my practice target.

4. **Connectivity Check:** Configured the virtual network adapters so both machines reside on the same internal subnet.

6. What I Learned

- **The Concept of Virtualization:** I learned how to run an entire OS as a "guest" without affecting my "host" (main) computer.
- **The Attacker vs. Victim Lab Model:** In ethical hacking, you always need a target. Metasploitable serves as a safe environment to test exploits without breaking the law.
- **Network Isolation:** I learned the importance of **Isolating the Lab**. Setting up a private network prevents accidental "scanning" of my home router, which is a crucial professional habit.
- **Snapshots:** I learned how to take a "Snapshot" in VirtualBox. If I accidentally delete a critical file while hacking, I can "teleport" the machine back to its original working state with one click.

Key Takeaway for Newbies: *"A good hacker is only as good as their lab. Setting this up correctly is the most important step to prevent legal issues and system crashes."*

7. Hands-on Examples & Verification

- **Command used to check IP:** `ifconfig`
  - *Usage:* I ran this on both machines to ensure they are on the same subnet (e.g., both starting with `192.168.56.x`).
- **Verification:** I successfully "pinged" the Metasploitable machine from my Kali terminal to confirm they are connected.

DAY 2

Mastering the Linux Terminal & Essential Commands

1. Introduction

In ethical hacking, the **Terminal (CLI)** is your most powerful weapon. While Windows and macOS rely heavily on a Graphical User Interface (GUI), a hacker performs 95% of their work using text-based commands. Today was all about moving away from the mouse and learning to talk directly to the operating system.

2. Objective of the Day

The goal of Day 2 is to become comfortable navigating the Linux file system, managing files and folders, and understanding the core commands required to run security tools in Kali Linux.

3. Why Linux for Hacking?

Linux is the preferred OS for cybersecurity because:

- **Open Source:** You can modify any part of the system.
- **Lightweight:** It runs efficiently on virtual machines.
- **Control:** Most hacking tools are built specifically for the Linux terminal.

4. Essential Command Categories

A. Navigation Commands

- `pwd` (Print Working Directory): Tells you exactly which folder you are in.
- `ls`: Lists all files and folders in the current directory.
    - `ls -la`: Shows hidden files and detailed info (permissions, size).
- `cd [folder]`: Change directory to move between folders.
    - `cd ..`: Moves you one step back (to the parent folder).

B. File & Directory Management

- `mkdir [name]`: Creates a new folder.
- `touch [name].txt`: Creates an empty text file.
- `cp [source] [destination]`: Copies a file.
- `mv [source] [destination]`: Moves or renames a file.
- `rm [file]`: Deletes a file (**Be careful! There is no "Recycle Bin" in the terminal**).
- `cat [file]`: Displays the content of a file on the screen.

C. System & Network Info

- `whoami`: Tells you which user account you are currently using.
- `top`: Shows running processes and RAM usage (like Task Manager).

- `clear`: Cleans up your terminal screen.

5. User Permissions & Sudo

Linux is built on permissions.

- **Standard User:** Has limited access.
- **Root User:** The "Superuser" who can do anything.
- **Sudo (SuperUser Do):** We use this prefix before a command to run it with administrative privileges (e.g., `sudo apt update`).

6. What I Did Today

1. **Terminal Immersion:** I spent the day exclusively using the terminal in Kali Linux instead of the file explorer.
2. **Folder Architecture:** I created a structured workspace for my internship tasks using `mkdir`.
3. **File Manipulation:** I practiced creating, reading, and moving logs and notes using `touch`, `cat`, and `mv`.
4. **Searching:** I used the `grep` command to search for specific text within a file, a vital skill for analyzing large log files later on.

7. What I Learned

- **Speed & Efficiency:** I realized that typing a command is often much faster than clicking through five different folders.
- **Case Sensitivity:** Linux is case-sensitive. `Folder1` and `folder1` are two completely different things!
- **Tab Completion:** I learned the "Tab" key trick. Typing the first few letters of a command and hitting Tab completes it automatically—this saves time and prevents typos.
- **The Power of Man Pages:** I learned that typing `man [command]` (e.g., `man nmap`) opens a full manual for any tool.

Key Takeaway for Newbies:

"Don't be afraid of the black screen. The terminal isn't just a place to type codes; it's a direct conversation with the computer's brain."

8. Hands-on Examples & Practice

- **Task:** Create a folder named "Internship_Notes", go inside it, and create a file named "Day2.txt".

Execution:

1. `mkdir Internship_Notes`
2. `cd Internship_Notes`
3. `touch Day2.txt`
4. `ls` (to verify the file exists)

```
┌──(kali㉿kali)-[~]
└─$ pwd
/home/kali

┌──(kali㉿kali)-[~]
└─$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos

┌──(kali㉿kali)-[~]
└─$ cd Desktop

┌──(kali㉿kali)-[~/Desktop]
└─$ mkdir Internship_Notes

┌──(kali㉿kali)-[~/Desktop]
└─$ ls
'Guest OS'   Internship_Notes

┌──(kali㉿kali)-[~/Desktop]
└─$ cd Internship_Notes

┌──(kali㉿kali)-[~/Desktop/Internship_Notes]
└─$ echo "Day 2 complete" > Day2.txt

┌──(kali㉿kali)-[~/Desktop/Internship_Notes]
└─$ ls
Day2.txt

┌──(kali㉿kali)-[~/Desktop/Internship_Notes]
└─$ cat Day2.txt
Day 2 complete

┌──(kali㉿kali)-[~/Desktop/Internship_Notes]
└─$
```

DAY 3

Networking Foundations & Wireshark Analysis

1. Introduction

If hacking is the art of entry, networking is the map of the building. To find vulnerabilities in a system, you must first understand how data moves from Point A to Point B. Today was dedicated to stripping away the user interface and looking at the raw pulses of data **packets** that travel across a wire.

2. Objective of the Day

The goal of Day 3 is to master the fundamentals of communication protocols and learn how to use a packet sniffer (Wireshark) to "eavesdrop" on network traffic in our secure lab environment.

3. Core Networking Concepts

A. The TCP/IP Model

Everything on the internet follows this 4-layer structure:

- **Application Layer:** Where your browser (HTTP) or email (SMTP) lives.
- **Transport Layer:** Ensures data arrives correctly (TCP/UDP).
- **Internet Layer:** Routes packets using IP addresses.
- **Network Access Layer:** Handles physical hardware (Ethernet/Wi-Fi).

B. IP vs. MAC Address

- **IP Address (Logical):** Like your house address; it can change if you move (e.g., `192.168.1.10`).
- **MAC Address (Physical):** Like your Social Security Number; it is burned into your hardware at the factory.

C. Common Ports (The "Doors")

Port	Protocol	Service
21	FTP	File Transfer
22	SSH	Secure Remote Login
80	HTTP	Unsecured Web Traffic
443	HTTPS	Secured Web Traffic

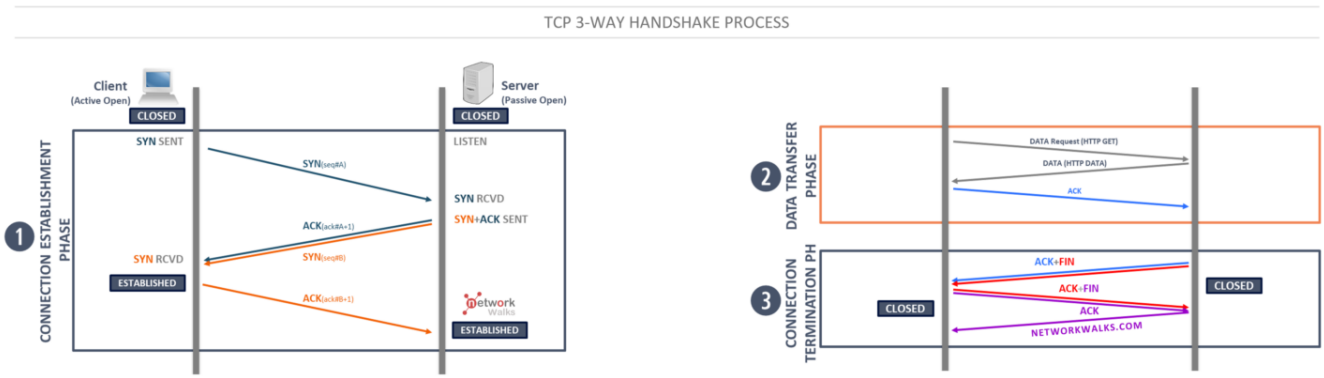
4. Tools Used: Wireshark

Wireshark is a "Packet Sniffer." It captures every piece of data entering or leaving your network card and translates it into a format humans can read.

- **Promiscuous Mode:** Allows the network card to see all traffic on the network, not just traffic intended for it.
- **Filters:** Help us find specific things (e.g., typing `http` in the bar shows only web traffic).

5. What I Did Today

1. **Protocol Analysis:** I studied the difference between **TCP** (reliable, requires a handshake) and **UDP** (fast, but no confirmation of receipt).
2. **The Three-Way Handshake:** I used Wireshark to observe the **SYN -> SYN-ACK -> ACK** process that happens every time a connection is made.
3. **Lab Traffic Capture:** I started a capture in Kali, then pinged my Metasploitable machine. I watched the **ICMP** (Ping) packets appear in real-time.
4. **Network Tools:** I practiced using `netstat -rn` to see my routing table and `nslookup` to see how domain names turn into IPs.



6. What I Learned

- **Everything is Visible:** I realized that on an unencrypted connection (like HTTP or FTP), passwords can be seen in plain text through Wireshark.
- **Protocol Hierarchy:** Understanding that a packet is like an envelope inside an envelope; each layer adds its own "header" info.
- **Troubleshooting:** I learned that `traceroute` is vital for seeing where a connection is failing between multiple routers.

Key Takeaway for Newbies:

"Networking isn't just about connecting computers; it's about understanding the language they speak. If you know the language, you can hear the secrets."

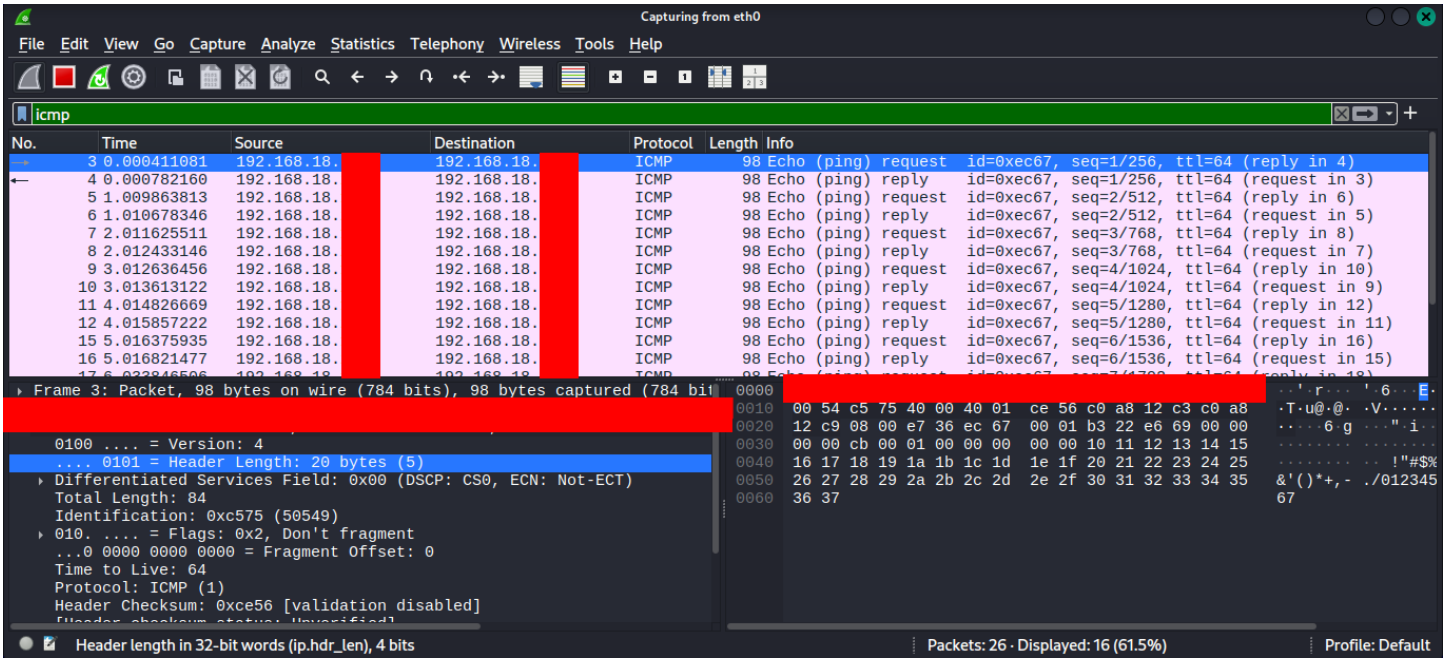
7. Hands-on Examples & Practice

- **Task:** Identify the IP of your lab gateway and capture a "Ping" request in Wireshark.

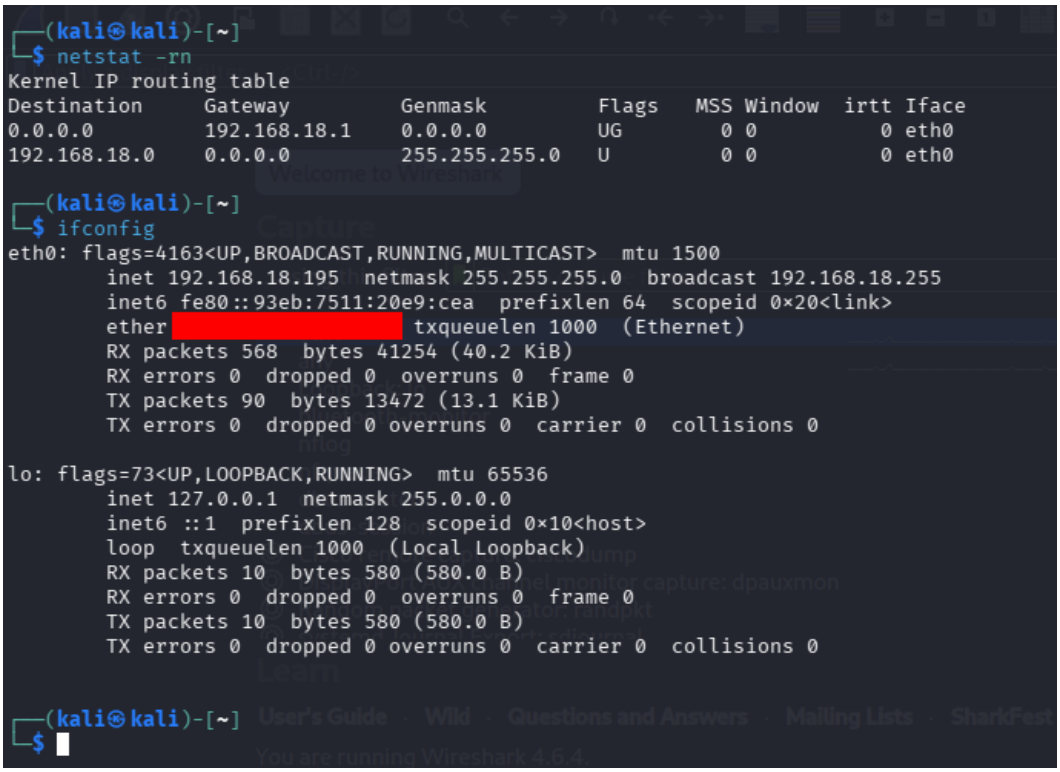
Execution:

1. Open Wireshark and start capturing on `eth0`.
2. Open Terminal and type: `ping -c 4 [Metasploitable-IP]`.
3. Go back to Wireshark and type `icmp` in the filter bar.
4. Observe the "Request" and "Reply" packets.

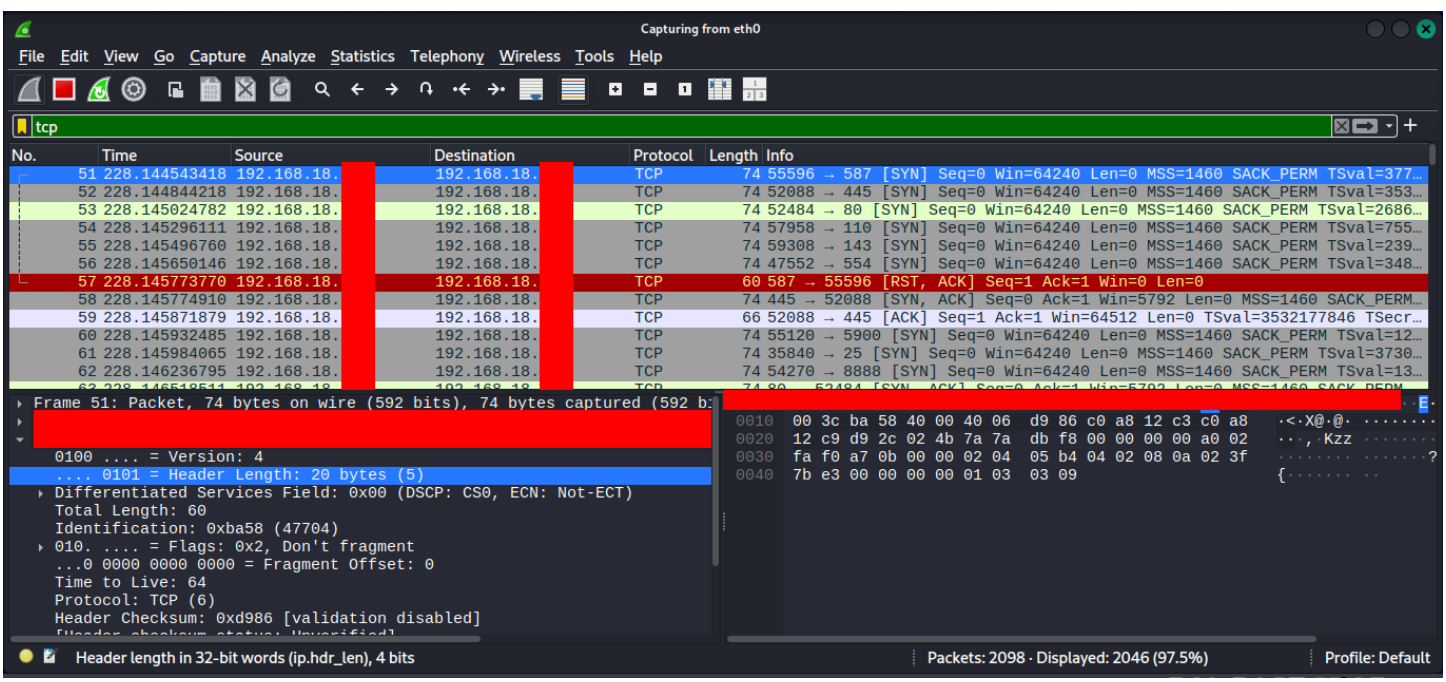
Screenshot 1: Wireshark running with the `icmp` filter active, showing the ping packets.



Screenshot: A terminal window showing the output of `netstat` or `ifconfig`.



Screenshot 3: A close-up of a TCP packet in Wireshark showing the "Source" and "Destination" ports.



DAY 4

Footprinting & Reconnaissance (Passive)

1. Introduction

In the world of ethical hacking, "Footprinting" is the first step of an attack. It is the process of gathering as much information as possible about a target network to identify potential entry points. Today, we focused on **Passive Reconnaissance**, which means collecting data without ever interacting directly with the target's servers.

2. Objective of the Day

The goal of Day 4 is to learn how to use publicly available resources (OSINT - Open Source Intelligence) to map out a target's digital footprint, including domain ownership, IP ranges, and employee information.

3. The Reconnaissance Phase

A. Passive vs. Active Recon

- **Passive Recon:** You gather info from 3rd party sources (Google, LinkedIn, WHOIS). The target has **no way** of knowing you are looking at them.
- **Active Recon:** You interact with the target (Port scanning, pinging). This is faster but can be logged by firewalls and security systems.

B. What are we looking for?

- **Domain Details:** Who owns the site? When does it expire?
- **IP Addresses:** What is the server's actual location?
- **Email Addresses:** Useful for future social engineering or credential testing.
- **Subdomains:** Often, subdomains like `dev.example.com` or `test.example.com` are less secure than the main site.

4. Tools & Techniques Used

1. **WHOIS:** A database that contains the registration information of domain names.
2. **NSLookup / Dig:** Used to query DNS servers to find IP addresses and mail server records.
3. **TheHarvester:** A powerful Kali tool that scrapes search engines and social media for emails, subdomains, and hostnames.
4. **Shodan.io:** Known as the "Search Engine for the Internet of Things (IoT)." It finds connected devices like webcams, industrial routers, and servers.

5. What I Did Today

1. **Domain Research:** I performed a `whois` lookup on a public domain to identify the registrar and name servers.

2. **Email Harvesting:** I used `theHarvester` to see how much public information is available about a sample organization.
3. **DNS Enumeration:** I used `nslookup` to find the Mail Exchange (MX) records of a target, showing where their emails are processed.
4. **Google Dorking:** I practiced using advanced search operators (like `site:`, `filetype:pdf`, `intitle:index of`) to find exposed sensitive files on the internet.

6. What I Learned

- **Information Leakage:** It is surprising how much information a company "leaks" just by existing online.
- **The Importance of Privacy:** I learned why companies use "WHOIS Protection" to hide their personal contact details from hackers.
- **OSINT Power:** You can find almost everything about a target without ever sending a single packet to their server.

Key Takeaway for Newbies:

"Reconnaissance is 70% of a successful hack. The more you know about your target, the easier it is to find a way in."

7. Hands-on Examples & Intelligence Gathering

- **Task:** Use the terminal to find the owner of a domain and its IP address.

Execution:

1. `whois google.com` (To see registration details).
2. `nslookup google.com` (To see the IP addresses).
3. `theHarvester -d example.com -b crtsh -l 50` (To search for emails).

    - To use this tool, we will need API keys, which we will save in this location.
        - `“nano etc/theHarvester/api-keys.yaml”`

Screenshot 1: Terminal showing the output of a `whois` command.

```
┌──(kali㉿kali)-[~]
└─$ whois faceboo.com
   Domain Name: FACEBOO.COM
   Registry Domain ID: 132537079_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois.registrarsec.com
   Registrar URL: http://www.registrarsec.com
   Updated Date: 2025-07-14T09:00:12Z
   Creation Date: 2004-10-11T21:51:00Z
   Registry Expiry Date: 2026-10-11T21:51:00Z
   Registrar: RegistrarSEC LLC
   Registrar IANA ID: 2475
   Registrar Abuse Contact Email: abusecomplaints@registrarsec.com
   Registrar Abuse Contact Phone: +1-650-308-7004
   Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
   Name Server: A.NS.FACEBOOK.COM
   Name Server: B.NS.FACEBOOK.COM
   Name Server: C.NS.FACEBOOK.COM
   Name Server: D.NS.FACEBOOK.COM
   DNSSEC: unsigned
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-04-20T13:46:51Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar.  Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
```

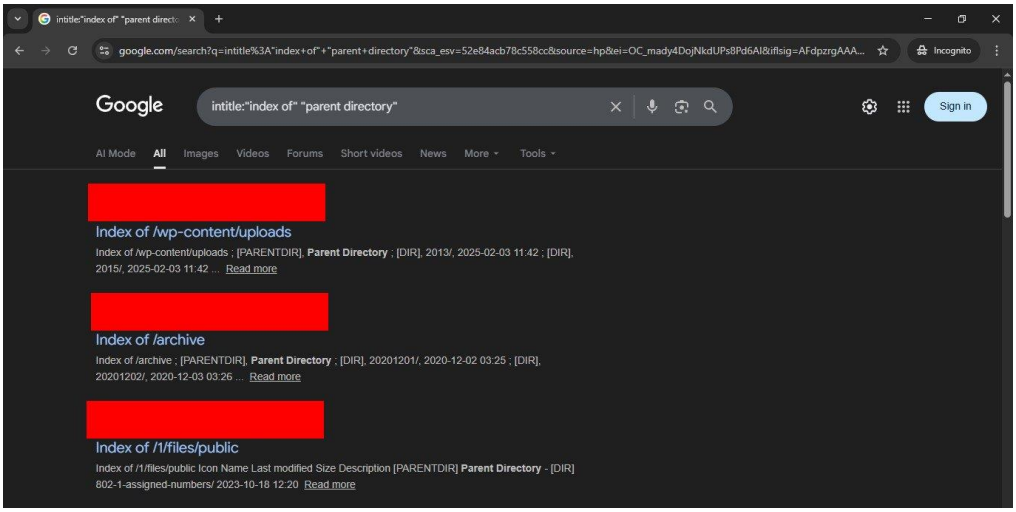
Screenshot 2: Results of an `nslookup` query showing "Non-authoritative answer."

```
┌──(kali㉿kali)-[~]
└─$ nslookup facebook.com
Server:
Address:

Non-authoritative answer:
Name:   facebook.com
Address: 57.144.148.1
Name:   facebook.com
Address: 2a03:2880:f34a:1:face:b00c:0:25de


┌──(kali㉿kali)-[~]
└─$
```

Screenshot 3: A "Google Dork" search result (e.g., searching for `filetype:log` for a specific site).



DAY 5

Active Reconnaissance & Nmap Scanning

1. Introduction

After gathering public information in Day 4, it was time to "touch" the target. Today, we moved into **Active Reconnaissance**. This phase involves interacting directly with the target machine to see which ports are open, which services are running, and what operating system is being used.

2. Objective of the Day

The goal of Day 5 is to master **Nmap**, the industry-standard tool for network discovery. I focused on identifying open "doors" (ports) on the Metasploitable2 machine and understanding how to read scan results to find potential vulnerabilities.

3. Understanding Nmap (Network Mapper)

The Three States of a Port:

- **Open:** The service is actively listening for connections (e.g., a web server on port 80).
- **Closed:** The port is reachable but no service is running on it.
- **Filtered:** A firewall is blocking Nmap from determining if the port is open or closed.

4. Types of Scans

A. Stealth Scan (-sS) vs. TCP Connect (-sT)

- **TCP Connect Scan (-sT):** Completes the full "Three-Way Handshake." It is very accurate but very easy for firewalls to log.
- **SYN Stealth Scan (-sS):** This is the default. It sends a SYN packet but breaks the connection before it's finished. This often avoids triggering basic logs.

B. Version & OS Detection

- **-sV:** Probes open ports to determine what software and **version** is running.
- **-O:** Attempts to guess the **Operating System** (Linux, Windows, etc.) by analyzing how the stack responds to packets.

5. What I Did Today

- **Host Discovery:** I used `nmap -sn [Network/24]` to find all live devices in my virtual lab.
- **Service Enumeration:** I ran a version scan (`-sV`) on the Metasploitable2 IP.

- **Aggressive Scanning:** I practiced using the `-A` flag, which combines OS detection, version detection, script scanning, and traceroute in one command.
- **Output Management:** I learned how to save my results into a file using `-oN` (Normal format) so I can review them later without re-scanning.

6. What I Learned

- **Service Vulnerability:** I learned that an open port isn't a vulnerability by itself, but the **version** of the service running on it might be (e.g., an old version of FTP).

- **Timing is Everything:** Nmap has timing templates (-T0 to -T5). I learned that -T4 is great for lab environments, while -T1 or -T2 is used to stay "sneaky" on real networks.

- **The Power of Scripts:** I discovered the Nmap Scripting Engine (NSE), which allows Nmap to do more than just scan it can check for specific bugs automatically.

Key Takeaway for Newbies:

"Nmap is the eyes of a hacker. Without it, you are attacking in the dark."

7. Hands-on Examples: Scanning the Lab

- **Task:** Perform a full service and OS detection scan on the victim machine.

Execution:

1. `nmap -sV -O [Victim-IP]`
2. Check the "STATE" and "SERVICE" columns.
3. Identify the version of the "vsftpd" service.

Screenshot 1: Your terminal showing the output of an Nmap scan (e.g., `nmap -sV [IP]`).

```
┌──(kali㉿kali)-[~/Desktop/Internship_Notes]
└─$ nmap -sV -O [redacted]
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-21 03:06 -0400
Nmap scan report for [redacted]
Host is up (0.00092s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login       OpenBSD or Solaris rlogind
514/tcp  open  tcpwrapped
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
MAC Address: [redacted] (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.74 seconds
```

Screenshot 2: A screenshot showing the OS detection results (e.g., "Running: Linux 2.6.X").

```
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
```

DAY 6

Vulnerability Analysis (Finding the Cracks)

1. Introduction

Reconnaissance told us which doors are open (Ports) and who is standing behind them (Services). Today, we move to the next logical step: **Vulnerability Analysis**. This is the process of checking those services against databases of known bugs, misconfigurations, and weaknesses to see if there is a way to gain unauthorized access.

2. Objective of the Day

The goal of Day 6 is to bridge the gap between "Scanning" and "Exploitation." I focused on taking the version numbers discovered in Day 5 and researching if they are "vulnerable." I also explored how to use automated scripts to find common security flaws.

3. What is Vulnerability Analysis?

Vulnerability analysis is the systematic review of security weaknesses in an information system. In our lab, we are looking for:

- **Unpatched Software:** Services like `vsftpd 2.3.4` that have famous "backdoors."
- **Default Credentials:** Systems that still use `admin/admin` or `root/root`.
- **Misconfigurations:** Services that allow anonymous login when they shouldn't.

4. Manual vs. Automated Analysis

- **Manual Research:** Using the service name and version number (e.g., *Apache 2.2.8*) and searching it on **Exploit-DB** or **Google**.
- **Automated Scanning:** Using tools that automatically check for thousands of vulnerabilities at once.

5. The Nmap Scripting Engine (NSE)

Nmap is not just a scanner; it's a vulnerability hunter. The **NSE** allows us to run scripts that check for specific vulnerabilities.

- **Auth scripts:** Check for default passwords.
- **Vuln scripts:** Check for specific known exploits.
- **Default scripts (-sC):** Runs a set of safe, common scripts.

6. What I Did Today

1. **Vulnerability Research:** I took the service list from my Day 5 Nmap scan and used the `searchsploit` command in Kali to see if any exploits exist for those versions.
2. **Running Vuln Scripts:** I ran an Nmap vulnerability scan using the command:

```
nmap --script vuln [Victim-IP]
```

1. **Analyzing Results:** I identified that the FTP service on Metasploitable2 is highly vulnerable to a specific "Backdoor" attack.
2. **Documentation:** I mapped each open port to a potential risk, creating a "Vulnerability Map" of my victim machine.

7. What I Learned

- **Versions Matter:** I learned that even a small change in a version number (like 2.3.4 vs 2.3.5) can be the difference between a secure system and a completely hacked one.
- **False Positives:** Sometimes automated tools say a system is vulnerable when it isn't. I learned to verify my findings with manual research.
- **CVEs:** I learned about **Common Vulnerabilities and Exposures (CVE)**—the unique ID numbers given to every major security bug in the world.

Key Takeaway for Newbies:

"Don't just scan; analyze. A scanner tells you a window is open; analysis tells you if the lock is broken."

8. Hands-on Examples: Identifying Vulnerabilities

- **Task:** Use Nmap scripts to find a vulnerability on the Metasploitable2 machine.

Execution:

1. Run: `nmap --script ftp-vsftpd-backdoor -p 21 [Victim-IP]`
2. Observe the output. If it says "VULNERABLE," you have found a way in!
3. Use `searchsploit vsftpd` to see the details of the exploit.

Screenshot 1: Terminal showing the results of `nmap --script vuln`.

```
┌──(kali㉿kali)-[~/Desktop/Internship_Notes]
└─$ nmap --script ftp-vsftpd-backdoor -p 21[redacted]
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-21 03:19 -0400
Nmap scan report for[redacted]
Host is up (0.00082s latency).

PORT   STATE SERVICE
21/tcp open  ftp
| ftp-vsftpd-backdoor:
|   VULNERABLE:
|   vsFTPd version 2.3.4 backdoor
|     State: VULNERABLE (Exploitable)
|     IDs:  CVE:CVE-2011-2523  BID:48539
|       vsFTPd version 2.3.4 backdoor, this was reported on 2011-07-04.
|     Disclosure date: 2011-07-03
|     Exploit results:
|       Shell command: id
|       Results: uid=0(root) gid=0(root)
|     References:
|       https://www.securityfocus.com/bid/48539
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
|       https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
|_      http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html
MAC Address:[redacted](Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.56 seconds

┌──(kali㉿kali)-[~/Desktop/Internship_Notes]
└─$
```

Screenshot 2: The output of the `searchsploit` command showing a list of exploits.

```
┌──(kali㉿kali)-[~/Desktop/Internship_Notes]
└─$ searchsploit vsftpd
------------------------------------------------------------------------------------------ ---------------------------------
 Exploit Title                                                                            |  Path
------------------------------------------------------------------------------------------ ---------------------------------
vsftpd 2.0.5 - 'CWD' (Authenticated) Remote Memory Consumption                            | linux/dos/5814.pl
vsftpd 2.0.5 - 'deny_file' Option Remote Denial of Service (1)                            | windows/dos/31818.sh
vsftpd 2.0.5 - 'deny_file' Option Remote Denial of Service (2)                            | windows/dos/31819.pl
vsftpd 2.3.2 - Denial of Service                                                          | linux/dos/16270.c
vsftpd 2.3.4 - Backdoor Command Execution                                                 | unix/remote/49757.py
vsftpd 2.3.4 - Backdoor Command Execution (Metasploit)                                    | unix/remote/17491.rb
vsftpd 3.0.3 - Remote Denial of Service                                                   | multiple/remote/49719.py
------------------------------------------------------------------------------------------ ---------------------------------
Shellcodes: No Results

┌──(kali㉿kali)-[~/Desktop/Internship_Notes]
└─$ searchsploit vsftpd 2.3.4
------------------------------------------------------------------------------------------ ---------------------------------
 Exploit Title                                                                            |  Path
------------------------------------------------------------------------------------------ ---------------------------------
vsftpd 2.3.4 - Backdoor Command Execution                                                 | unix/remote/49757.py
vsftpd 2.3.4 - Backdoor Command Execution (Metasploit)                                    | unix/remote/17491.rb
------------------------------------------------------------------------------------------ ---------------------------------
Shellcodes: No Results
```

Screenshot 3: A screenshot of **Exploit-DB** or a CVE website showing details of a vulnerability you found.

EXPLOIT DATABASE

vsftpd 2.3.4 - Backdoor Command Execution

EDB-ID:	CVE:	Author:	Type:	Platform:	Date:
49757	2011-2523	HERCULESRD	REMOTE	UNIX	2021-04-12

EDB Verified: ✓ Exploit: / {} Vulnerable App:

```
# Exploit Title: vsftpd 2.3.4 - Backdoor Command Execution
# Date: 9-04-2021
# Exploit Author: HerculesRD
# Software Link: http://www.linuxfromscratch.org/~thomasp/blfs-book-xsl/server/vsftpd.html
# Version: vsftpd 2.3.4
# Tested on: debian
# CVE : CVE-2011-2523

#!/usr/bin/python3

from telnetlib import Telnet
import argparse
from signal import signal, SIGINT
from sys import exit
```

DAY 7

Final Lab Review & Internship Week 1 Complete

1. Introduction

The first week of the **YoungDevInterns Ethical Hacking Internship** has come to an end. We have moved from absolute zero having no environment to being able to map out and analyze the vulnerabilities of a remote target. Today is about "The Big Picture": reviewing our methodology and preparing a professional report of our findings.

2. Objective of the Day

The goal of Day 7 is to perform a final audit of the virtual lab, ensure all tools are working correctly, and summarize the technical progress made during the **Foundations of Ethical Hacking** week.

3. The Ethical Hacking Methodology

During this week, we followed the professional standards used by penetration testers worldwide:

1. **Reconnaissance (Passive):** Gathering info without touching the target.
2. **Scanning (Active):** Finding open ports and services.
3. **Vulnerability Analysis:** Identifying weaknesses in those services.

4. Final Lab Audit

Before finishing the week, I verified that my lab environment is optimized for Week 2 (which usually involves Exploitation).

- **Snapshots Check:** I ensured I have a "Clean State" snapshot of both Kali and Metasploitable2.
- **Network Integrity:** I confirmed that the machines are still isolated from my host network but can communicate with each other.

5. Documentation & Reporting

In the professional world, a hacker is only as good as their report. Today, I compiled all my terminal logs and screenshots into a structured format. This documentation serves as the "Proof of Work" for the internship.

6. What I Did Today

1. **Lab Review:** I re-ran the core commands from the week (`ifconfig`, `nmap`, `whois`) to ensure I have mastered the syntax.
2. **Report Finalization:** I organized my daily notes into this final Week 1 document for submission.
3. **Github Management:** I pushed my documentation and task files to my repository: **YoungDevInterns_EthicalHacking_Tasks**.
4. **Social Media Update:** I prepared my LinkedIn and Facebook posts to share my progress with the community.

7. What I Learned

- **Consistency is Key:** I learned that hacking isn't about one "magic button"; it's a step-by-step process of elimination.
- **The Value of Logs:** I realized how important it is to save Nmap results (`-oN`) because you cannot remember every port and version number.
- **Legal Boundaries:** Most importantly, I solidified my understanding of the legal and ethical boundaries of this profession.

Key Takeaway for Newbies:

"The hardest part of hacking isn't the code; it's the patience. This week proved that building a strong foundation makes the actual 'hacking' much easier."

8. Weekly Summary Checklist

Task	Status
✓ **Virtual Lab (Kali + Metasploitable) Setup**	✓ Complete
✓ **Linux Terminal Basic Commands**	✓ Mastered
✓ **Networking Protocols (TCP/IP, Ports)**	✓ Understood
✓ **Passive Recon (WHOIS, DNS)**	✓ Performed
✓ **Active Scanning (Nmap)**	✓ Performed
✓ **Vulnerability Identification**	✓ Identified